CSCE 763: Digital Image Processing
Homework #2

Instructor: Spring 2026
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Due: 1:15 pm EST, Tuesday, Feb 10

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Problem 1: Median Operator Nonlinearity (10 pts)

The median ζ, of a set of numbers is such that half the values in the set are below ζ, and the other half are above it. For example, the median of the set of values $\{12, 3, 8, 20, 21, 75, 31\}$ is 20. Show that an operator that computes the median of a subimage area, S, is nonlinear. **Hints:** an operator H is linear if

$$H[a_1 f_1(x, y) + a_2 f_2(x, y)] = a_1 H[f_1(x, y)] + a_2 H[f_2(x, y)]$$

Step 1: State the Linearity Condition

An operator H is linear if and only if the **superposition property** holds for all images f_1, f_2 and all scalars a_1, a_2:

$$H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$$

To show H is nonlinear, it suffices to find a single counterexample where this equality fails. We set $a_1 = a_2 = 1$ and test the additivity property over a 1×3 subimage region S.

Step 2: Define the Counterexample

Choose two images defined by their pixel values over S:

$$f_1 = \{2,\ 3,\ 10\}, \qquad f_2 = \{10,\ 1,\ 2\}$$

Step 3: Compute the Right-Hand Side

$$H[f_1] + H[f_2] = \text{median}\{2, 3, 10\} + \text{median}\{10, 1, 2\} = 3 + 2 = 5$$

Step 4: Compute the Left-Hand Side

$$f_1 + f_2 = \{2 + 10,\ 3 + 1,\ 10 + 2\} = \{12,\ 4,\ 12\}$$
$$H[f_1 + f_2] = \text{median}\{12, 4, 12\} = 12$$

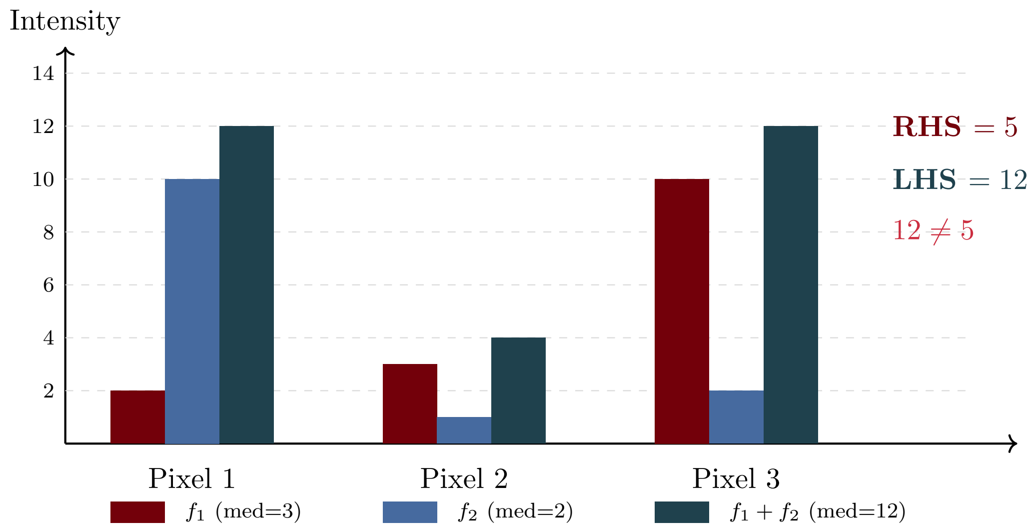
Step 5: Conclude

Since

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2]$$

the superposition property is violated. Therefore, the median operator is **nonlinear**. ■

Visualization: Counterexample Bar Chart



Results

The median operator is **nonlinear**. The counterexample $f_1 = \{2, 3, 10\}$, $f_2 = \{10, 1, 2\}$ yields:

$$H[f_1 + f_2] = 12 \neq 5 = H[f_1] + H[f_2] \qquad \blacksquare$$

Problem 2: Image Subtraction for Defect Detection (20 pts)

Image subtraction is used often in industrial applications for detecting missing components in product assembly. The approach is to store a "golden" image that corresponds to a correct assembly; this image is then subtracted from incoming images of the same product. Ideally, the differences would be zero if the new products are assembled correctly. Difference images for products with missing components would be nonzero in the area where they differ from the golden image. What conditions do you think have to be met in practice for this method to work? You need to list at least two major factors that affect the detection performance.

Demonstration: Golden Image Analysis

To demonstrate these factors concretely, I applied various perturbations to a "golden" reference image of an industrial product (a pharmaceutical tablet) and computed the resulting difference images. Figure 1 shows the golden reference image used for this analysis.

Figure 1: Golden reference image of a pharmaceutical tablet with "FF" marking, used as the baseline for defect detection analysis.

Factors Affecting Detection Performance

1. **Illumination Intensity.** Changes in overall brightness between the golden image and test image cause uniform intensity shifts across the entire frame. Even modest variations of ± 40 intensity levels produce large difference signals that can overwhelm subtle defect signatures. Controlling this requires fixed, calibrated light sources with stable power supplies.

2. **Spatial Translation.** If the product is not positioned identically to when the golden image was captured, even small shifts of 5 pixels create prominent edge artifacts in the difference image. Mechanical fixtures and sub-pixel alignment algorithms are essential to prevent positional mismatches from triggering false positives.

3. **Rotational Misalignment.** When the product rotates slightly relative to the golden reference, arc-shaped difference patterns emerge around curved features. A rotation of just $2°$ can produce detectable artifacts, necessitating precise alignment fixtures or software-based rotation correction.

4. **Scale and Zoom Variation.** Differences in camera-to-object distance or focal length settings cause the test image to appear slightly larger or smaller than the reference. A 3% scale change creates ring-like edge differences, requiring fixed working distances and locked zoom settings.

5. **Sensor Noise.** Random electronic noise in the camera sensor introduces speckle patterns throughout the difference image. Gaussian noise with $\sigma = 20$ can mask small defects entirely, making low-noise sensors and temporal averaging across multiple frames important countermeasures.

6. **Focus Blur.** When the test image is slightly out of focus compared to the sharp golden reference, edges become softer and produce haloed differences. Reliable detection requires auto-focus locks or telecentric lenses that maintain consistent focus across the field of view.

7. **Non-uniform Illumination.** Vignetting and uneven light distribution cause the corners and edges of images to appear darker than the center. This spatial intensity gradient creates systematic differences even when the product is identical, requiring flat-field correction or carefully designed uniform lighting.

8. **Contrast and Gain Variation.** Changes in camera gain settings or lighting intensity ratios alter the dynamic range of captured images. A 30% contrast shift compresses or expands intensity differences, so fixed camera settings with automatic gain control disabled are necessary.

9. **Color Temperature.** Shifts between warm and cool illumination alter the balance of color channels differently. If the golden image was captured under daylight-balanced lighting but production uses warmer LEDs, the RGB channels will show systematic offsets requiring fixed white balance settings.

10. **Perspective Distortion.** Changes in the viewing angle between captures cause trapezoidal warping of the product's appearance. This geometric distortion creates systematic misalignment that cannot be corrected by simple translation or rotation, requiring telecentric optics or rigidly fixed camera mounts.

11. **Exposure Variation.** Differences in shutter speed or ambient light levels cause nonlinear brightness shifts that affect highlights and shadows differently. Consistent detection requires fixed exposure settings and controlled ambient lighting to eliminate environmental variability.

Failure Modes Summary

Figure 2 shows a summary of six key failure modes applied to the golden image. Each column shows: (1) the perturbed test image, (2) the resulting difference image (brighter = larger difference), and (3) the false positives detected at a fixed threshold. Notice how environmental variations that leave the product physically unchanged still trigger significant detection artifacts.

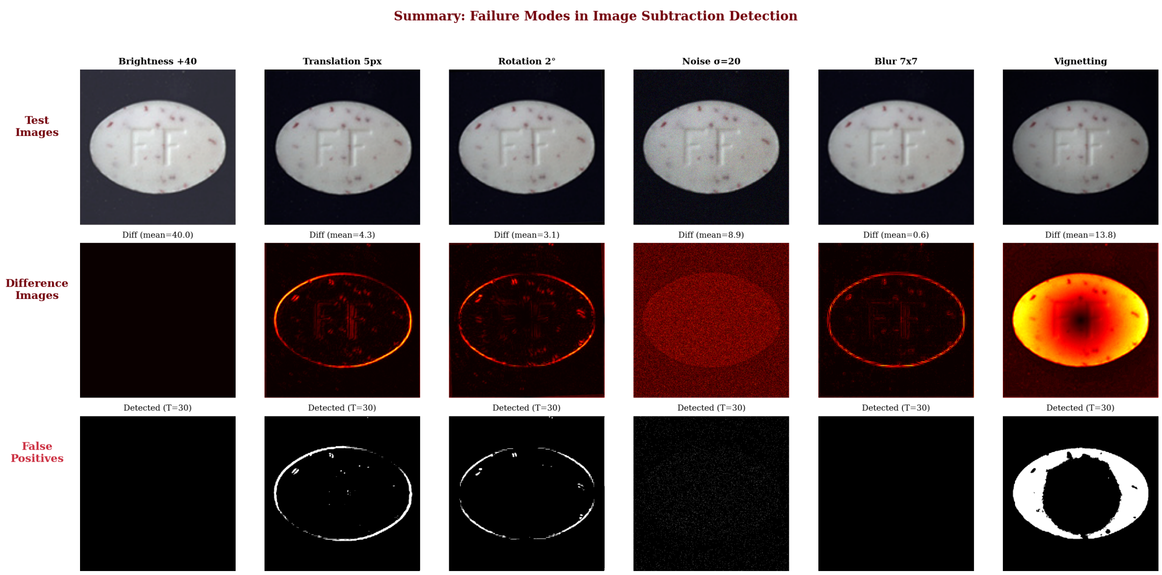


Figure 2: Summary of failure modes in image subtraction detection. Row 1: Test images with various perturbations. Row 2: Difference images (hotter = larger difference). Row 3: Detected regions at threshold $T = 30$. All perturbations produce false positives despite no actual defects.

Quantitative Comparison

Figure 3 quantitatively compares all twelve failure modes by mean absolute difference. The key insight is that **illumination-related factors dominate**: brightness changes and exposure variations cause the largest false positive signals. Geometric factors (translation, rotation, scale) cause smaller mean differences but concentrate at edges where defects are often located. Critically, a true defect (missing component) has a relatively **small** mean difference because it affects only a localized region—demonstrating how failure modes can mask real defects.

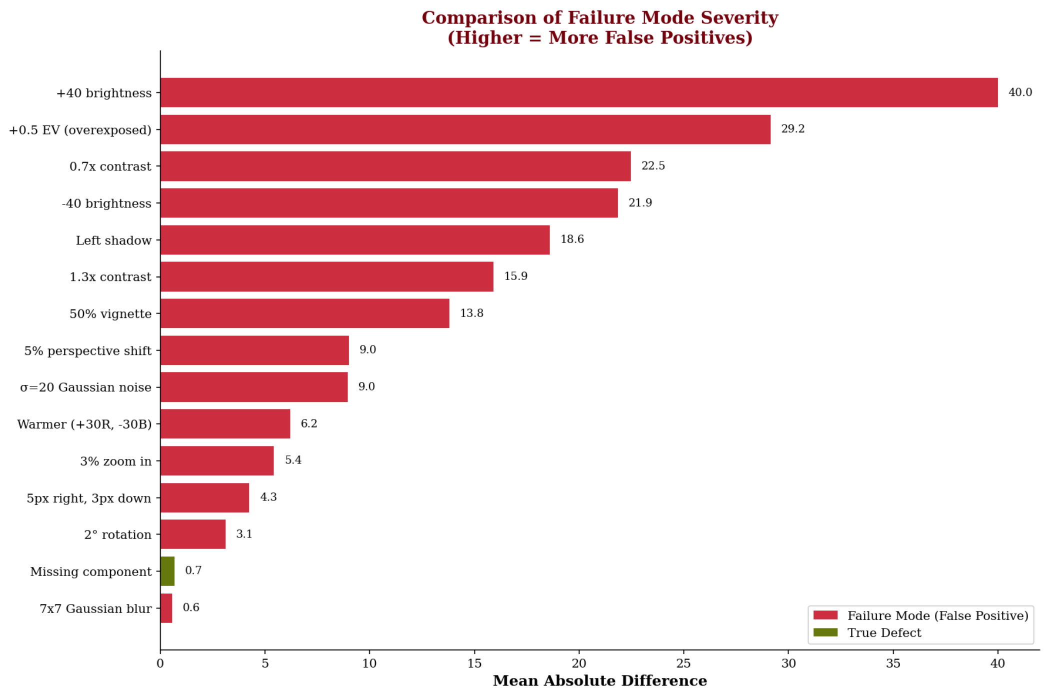


Figure 3: Comparison of failure mode severity measured by mean absolute difference. Higher values indicate more false positives. Note that the true defect (missing component, green) has a *lower* mean difference than most failure modes, illustrating how environmental variations can mask genuine defects.

Results

Major factors affecting image subtraction defect detection:

Primary Factors (Must Control):

1. **Illumination consistency:** Intensity, direction, uniformity, and color temperature must be identical.
2. **Spatial registration:** Position, orientation, and scale must match exactly.

Secondary Factors (Should Control):

3. Sensor noise (requires thresholding)
4. Focus/sharpness consistency
5. Camera gain and exposure settings
6. White balance / color calibration
7. Perspective and optical distortion
8. Shadow elimination

Problem 3: Histogram Equalization Idempotency (20 pts)

Suppose that a digital image is subjected to histogram equalization. Show that a second pass of histogram equalization (on the histogram-equalized image) will produce exactly the same result as the first pass.

Step 1: Define the Equalization Transform

Let the original image have intensity levels $r \in [0, L-1]$ with PDF $p_r(r)$ and CDF $P_r(r)$. The histogram equalization transformation is:

$$s = T(r) = (L-1)\,P_r(r) = (L-1)\int_0^r p_r(w)\,dw$$

Step 2: Show the Equalized Image Is Uniform

A fundamental result from probability theory: if $s = T(r) = (L-1)\,P_r(r)$, then s is **uniformly distributed** on $[0, L-1]$:

$$p_s(s) = \frac{1}{L-1}, \qquad 0 \le s \le L-1$$

Step 3: Apply Equalization Again

Apply histogram equalization a second time to the equalized image:

$$v = T_2(s) = (L-1)\,P_s(s) = (L-1)\int_0^s p_s(w)\,dw$$

Since $p_s(s) = \frac{1}{L-1}$, the CDF is:

$$P_s(s) = \int_0^s \frac{1}{L-1}\,dw = \frac{s}{L-1}$$

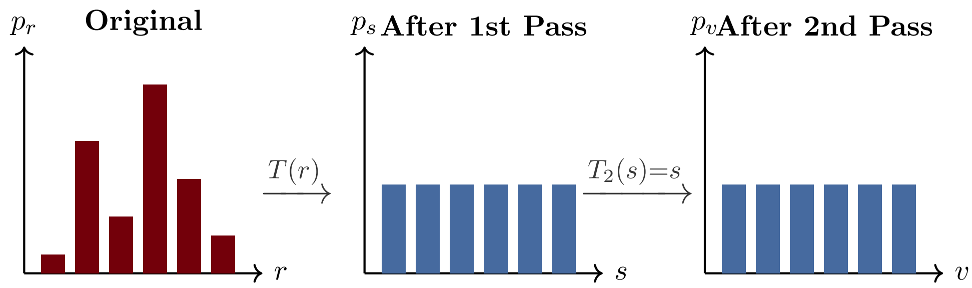
Step 4: Simplify to Identity

Substituting:

$$v = T_2(s) = (L-1)\cdot\frac{s}{L-1} = s$$

The second equalization is the **identity transformation**: $v = s$. The output is unchanged. ■

Visualization: Histogram Through Two Equalization Passes



Results

After the first histogram equalization, the output has a uniform distribution. Applying histogram equalization to a uniformly distributed image yields the identity mapping $v = s$. Therefore, the second pass produces exactly the same result as the first. ■

Problem 4: Histogram Matching (30 pts)

Suppose that a 4-bit image ($L = 16$) of size 64×64 has the intensity distribution as Table A. It is desired to transform the original histogram to a specified histogram as shown in Table B. What will be the actual histogram after the histogram matching operation?

r_k	n_k
0	16
1	0
2	512
3	32
4	64
5	128
6	64
7	1024
8	512
9	512
10	16
11	1024
12	0
13	128
14	0
15	64

Table A

z_q	n_q
0	64
1	128
2	512
3	512
4	1024
5	1024
6	512
7	128
8	128
9	64
10	0
11	0
12	0
13	0
14	0
15	0

Table B

Step 1: Setup

Total pixels: $N = 64 \times 64 = 4096$, intensity levels: $L = 16$, so $L - 1 = 15$.
The equalization transform is:

$$s_k = \text{round}\left(15 \cdot \sum_{j=0}^{k} \frac{n_j}{4096}\right)$$

Step 2: Cumulative Histogram and s_k Values from Table A

r_k	n_k	$\sum_{j=0}^{k} n_j$	$s_k = \text{round}\left(\frac{15 \cdot \sum n_j}{4096}\right)$
0	16	16	round(0.059) = 0
1	0	16	round(0.059) = 0
2	512	528	round(1.934) = 2
3	32	560	round(2.051) = 2
4	64	624	round(2.285) = 2
5	128	752	round(2.754) = 3
6	64	816	round(2.988) = 3
7	1024	1840	round(6.738) = 7
8	512	2352	round(8.613) = 9
9	512	2864	round(10.488) = 10
10	16	2880	round(10.547) = 11
11	1024	3904	round(14.297) = 14
12	0	3904	round(14.297) = 14
13	128	4032	round(14.766) = 15
14	0	4032	round(14.766) = 15
15	64	4096	round(15.000) = 15

Step 3: Cumulative Histogram and $G(z_q)$ Values from Table B

z_q	n_q	$\sum_{j=0}^{q} n_j$	$G(z_q) = \text{round}\left(\frac{15 \cdot \sum n_j}{4096}\right)$
0	64	64	round(0.234) = 0
1	128	192	round(0.703) = 1
2	512	704	round(2.578) = 3
3	512	1216	round(4.453) = 4
4	1024	2240	round(8.203) = 8
5	1024	3264	round(11.953) = 12
6	512	3776	round(13.828) = 14
7	128	3904	round(14.297) = 14
8	128	4032	round(14.766) = 15
9	64	4096	round(15.000) = 15
10	0	4096	15
11	0	4096	15
12	0	4096	15
13	0	4096	15
14	0	4096	15
15	0	4096	15

Step 4: Inverse Mapping $s_k \to z_q$

For each value of s_k, find the smallest z_q such that $G(z_q) \geq s_k$:

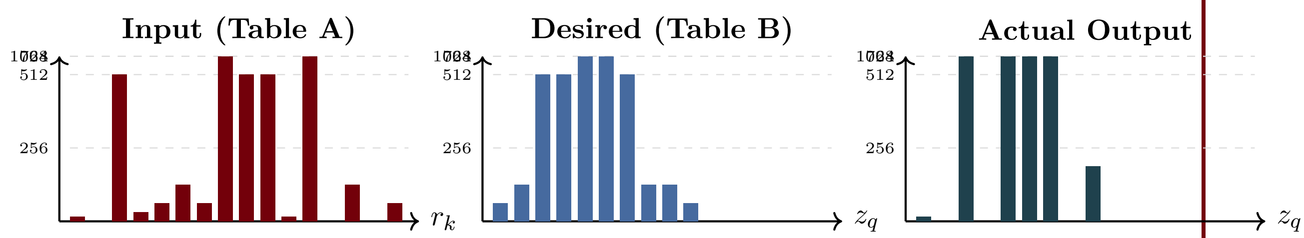
r_k	s_k	Smallest z_q with $G(z_q) \geq s_k$	z_q
0	0	$G(0) = 0 \geq 0$	0
1	0	$G(0) = 0 \geq 0$	0
2	2	$G(2) = 3 \geq 2$	2
3	2	$G(2) = 3 \geq 2$	2
4	2	$G(2) = 3 \geq 2$	2
5	3	$G(2) = 3 \geq 3$	2
6	3	$G(2) = 3 \geq 3$	2
7	7	$G(4) = 8 \geq 7$	4
8	9	$G(5) = 12 \geq 9$	5
9	10	$G(5) = 12 \geq 10$	5
10	11	$G(5) = 12 \geq 11$	5
11	14	$G(6) = 14 \geq 14$	6
12	14	$G(6) = 14 \geq 14$	6
13	15	$G(8) = 15 \geq 15$	8
14	15	$G(8) = 15 \geq 15$	8
15	15	$G(8) = 15 \geq 15$	8

Step 5: Build the Output Histogram

Using the mapping $r_k \to z_q$ and the pixel counts n_k from Table A:

Output z_q	Mapped from r_k	Pixel count
0	r_0, r_1	$16 + 0 = 16$
1	—	0
2	r_2, r_3, r_4, r_5, r_6	$512 + 32 + 64 + 128 + 64 = 800$
3	—	0
4	r_7	1024
5	r_8, r_9, r_{10}	$512 + 512 + 16 = 1040$
6	r_{11}, r_{12}	$1024 + 0 = 1024$
7	—	0
8	r_{13}, r_{14}, r_{15}	$128 + 0 + 64 = 192$
9–15	—	0 each
	Total:	$16 + 800 + 1024 + 1040 + 1024 + 192 = 4096$ ✓

Visualization: Input



Results

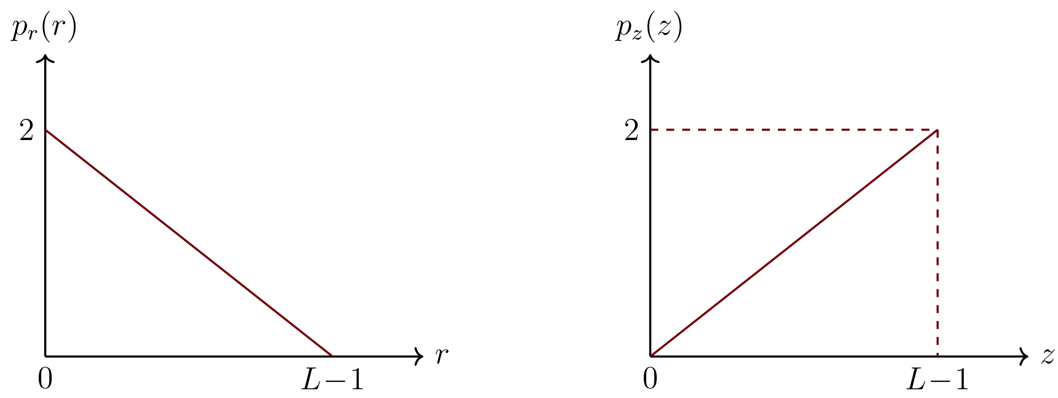
The actual output histogram after histogram matching is:

z_q	n_q^{out}
0	16
1	0
2	800
3	0
4	1024
5	1040
6	1024
7	0
8	192
9	0
10	0
11	0
12	0
13	0
14	0
15	0

The output histogram approximates the desired distribution but is not identical to it, due to the discrete nature of the intensity levels and the many-to-one mapping inherent in histogram matching.

Problem 5: Continuous Intensity Transformation (20 pts)

An image with intensities in the range $[0, L-1]$ has the pdf $p_r(r)$ shown in Figure (a). It is desired to transform the intensity levels of this image so that they will have the specified pdf $p_z(z)$ as shown in Figure (b). Assume continuous quantities and find the transformation $z = f(r)$ that will accomplish this.



(a) (b)

Step 1: Write $p_r(r)$ with Normalization Verification

Let $W = L - 1$. From Figure (a), $p_r(r)$ is a line from $(0, \frac{2}{W})$ to $(W, 0)$:

$$p_r(r) = \frac{2}{W}\left(1 - \frac{r}{W}\right) = \frac{2(W-r)}{W^2}, \qquad 0 \le r \le W$$

Verification: $\displaystyle\int_0^W \frac{2(W-r)}{W^2}\,dr = \frac{2}{W^2}\left[Wr - \frac{r^2}{2}\right]_0^W = \frac{2}{W^2}\cdot\frac{W^2}{2} = 1.$ ✓

Step 2: Write $p_z(z)$ with Normalization Verification

From Figure (b), $p_z(z)$ is a line from $(0, 0)$ to $(W, \frac{2}{W})$:

$$p_z(z) = \frac{2z}{W^2}, \qquad 0 \le z \le W$$

Verification: $\displaystyle\int_0^W \frac{2z}{W^2}\,dz = \frac{2}{W^2}\cdot\frac{W^2}{2} = 1.$ ✓

Step 3: Compute CDF $T(r)$

$$T(r) = \int_0^r p_r(w)\,dw = \int_0^r \frac{2(W-w)}{W^2}\,dw = \frac{2}{W^2}\left[Wr - \frac{r^2}{2}\right] = \frac{2Wr - r^2}{W^2}$$

Step 4: Compute CDF $G(z)$

$$G(z) = \int_0^z p_z(w)\,dw = \int_0^z \frac{2w}{W^2}\,dw = \frac{z^2}{W^2}$$

Step 5: Set $G(z)$

$$\frac{z^2}{W^2} = \frac{2Wr - r^2}{W^2}$$

$$z^2 = 2Wr - r^2 = r(2W - r)$$

$$\boxed{z = \sqrt{r(2(L-1) - r)}}$$

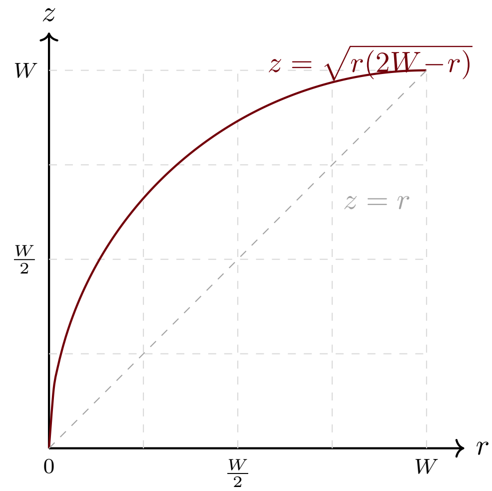
where we take the positive root since $0 \le r \le W$ ensures the argument is non-negative.

Step 6: Verify Boundary Conditions

- $f(0) = \sqrt{0 \cdot (2W - 0)} = 0$ ✓
- $f(W) = \sqrt{W(2W - W)} = \sqrt{W^2} = W = L - 1$ ✓

The transformation maps $[0, L-1] \to [0, L-1]$ as required.

Visualization: Transformation z



The curve is a quarter-circle arc from $(0,0)$ to (W,W), lying above the identity line, indicating the transformation boosts lower intensities.

Results

The transformation that maps the input PDF $p_r(r)$ to the desired output PDF $p_z(z)$ is:

$$z = f(r) = \sqrt{r\big(2(L-1)-r\big)}, \qquad 0 \le r \le L-1$$

This is a quarter-circle arc from $(0,0)$ to $(L-1,\ L-1)$. ■